REMEMBER FORWARD

TIER E · GRANITE VAULT

The permanent institutional archive.

$500–$1,500 · DESIGN TARGET 10,000+ YEARS · FIXED LOCATION

What Changed In This Edition

This edition removes the lead lining and the poured lead seal specified through August 2026, and withdraws the claim of "100,000+ years, the nuclear industry standard."

That claim was wrong. The nuclear industry does not use lead-lined stone for archival containment. Sweden and Finland's KBS-3 design uses a copper canister over a cast-iron insert, surrounded by a bentonite clay buffer, in granitic bedrock. Lead appears in that industry only as radiation shielding in transport casks. And the 100,000-year figure is claimed for repository GEOLOGY behind a formal safety case — never for a container, and never by us.

Lead is withdrawn on its own merits too: it creeps under sustained load so it cannot carry structure, it is toxic to cast and handle, and in the presence of moisture it forms a galvanic couple with titanium in which the lead is consumed. The molten-lead pouring step in earlier editions was also the most hazardous instruction in this entire project.

What replaces it is better and cheaper: a bentonite clay buffer and a sealed inner vessel.

Historical Precedent

Civilization	Application	Age	Survival Status
Ancient Egypt	Granite sarcophagi and canopic jars	3,000–5,000 years	Contents intact in many cases
Ancient Rome	Stone sarcophagi	2,000 years	Many survive intact with contents
Medieval Europe	Stone church crypts	500–1,000 years	Most survive structurally intact
Modern nuclear industry	Copper canister in a bentonite buffer, in granitic bedrock (KBS-3)	Designed for 100,000 years	Engineering standard, not time-tested. Their safety case, not ours.
This project	Civilizational knowledge archive	Design target: 10,000 years	Granite body, bentonite buffer, sealed inner vessel

Why Granite

Chemical properties

Granite is an igneous rock of quartz, feldspar and mica. It is chemically inert to virtually everything it will meet underground, it does not dissolve in groundwater on any timescale that matters here, and it is the host rock class that real deep repositories are actually built in.

Mechanical properties

Compressive strength of roughly 100 to 300 MPa. It will not be crushed by soil load, it will not deform, and unlike lead it does not creep. Granite carries the structure; nothing else in this design has to.

The Containment System — Three Layers

Layer 1 — the granite body

Structural and inert. It resists load, roots, burrowing animals and casual disturbance. It is not, by itself, a seal — stone joints are never hermetic.

Layer 2 — bentonite clay buffer

This is the layer that does the work the lead was imagined to do, and it does it better. Sodium bentonite swells roughly ten to fifteen times its dry volume when it meets water, which means any crack, gap or void it can reach becomes self-sealing. It buffers the chemistry around the inner vessel and slows water movement to a crawl. It is the engineered barrier in every serious geological repository.

Sourcing: sold as pond sealer, as drilling mud, and as unscented clumping cat litter. If you use cat litter, confirm it is sodium bentonite with no added fragrance, deodoriser or clumping agents.

Layer 3 — the sealed inner vessel

The only real barrier. Everything else buys it time.

- BUILD IT YOURSELF: a fired ceramic vessel sealed with hot pine pitch. In this project's own ranking of seals, pitch on ceramic rates equal to cast lead — and it is non-toxic, needs no crucible, and is the same method as the Tier 0 capsule.
- COMMISSION IT: a welded titanium vessel, if an institution can have one made. No gasket, no joint compound, nothing to fail. Titanium welding needs argon shielding and back-purge; this is a fabrication-shop operation, not a workshop one. Do not attempt it with a hardware-store welder.
- DO NOT USE any gasket, O-ring or elastomer, whatever the datasheet claims. No elastomer has a validated multi-century life. A 10,000-year vault with a 100-year seal is a 100-year vault.

Construction

Step 1 — obtain or cut the granite

Six slabs: base, four sides, lid. Minimum 5 cm thick, ideally 8 to 10 cm. Monument yards and stone fabricators sell offcuts cheaply. Alternatively carve a cavity into a single block — slower, and the traditional method, with no joints to fail.

Step 2 — assemble the box

Dry-fit all six slabs and grind any gaps with a diamond disc. Bed the joints in lime mortar. Geopolymer — local clay or metakaolin with an alkali activator, cured at low temperature — is an alternative worth considering: durable, low-permeability, and makeable from local materials by anyone who finds this later.

Step 3 — prepare and seal the inner vessel

- Dry the plates and any other contents thoroughly. Moisture sealed in is moisture that stays in.

- Load the vessel with desiccant and oxygen absorbers. Oxygen is the main agent of decay once water is excluded.
- Seal it: pitch on ceramic, or a welded titanium lid. Seal it permanently. The finder will cut it open.

Step 4 — pack with bentonite

Set the sealed vessel centrally in the granite cavity and pack dry sodium bentonite firmly all around it — a minimum of 5 cm on every face, including beneath. Leave no voids. Pack it dry: it is meant to swell later, when and if water ever arrives.

Step 5 — close and coat

Bed the lid on lime mortar and point the perimeter joint. When cured, coat the whole exterior in three or four layers of hot pine pitch or bitumen. This is a water-shedding layer, not a seal — it buys the mortar joints decades they would not otherwise have.

Burial and Placement

A vault of these dimensions weighs roughly 60 to 90 kg before contents. Plan the lift and the route before you build it, not after.

Placement	Depth / Location	Best For	Notes
Deep burial	Minimum 6 feet, ideally 8–10	Permanent hidden archive	Store GPS coordinates separately, in more than one place
Cave or rock shelter	On a raised shelf above floor level	Long-term accessible archive	Dry caves are the best preservation record we have
Foundation burial	Within a stone or concrete foundation	Community institutional archive	Protected by the structure above; the most accessible option
Arctic / permafrost	8–10 feet in a permafrost region	NO LONGER RECOMMENDED ALONE	Permafrost is not a permanent assumption. Svalbard's seed vault took meltwater into its access tunnel within nine years of opening. Treat frozen ground as a bonus, never as the barrier.
Salt mine	On a raised platform above the mine floor	Excellent long-term archive	Self-sealing, dry, low oxygen. Requires an institutional agreement.

The Community Commitment

A granite vault is not just a container. It is heavy enough that it will not be moved casually, and permanent enough that it outlives whoever sites it. That makes siting it a decision about a place and the people in it, not a decision about a box.

Consider inscribing on the exterior: the year of burial, a brief plain description of the contents, and the bridge phrase — "This was made for you, freely, by people who remembered forward." Cut it deep. Raised letters abrade away first; a cut groove fills with dirt and gets easier to read, not harder.

Remember Forward · The Patient Message

Buy it. Build it. Bury it. For someone you will never meet.